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# Dissolved organic carbon enhances terrestrial carbon pathways with limited effects on freshwater fish trophic position

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## Data files

|                                                           |         |                       |
|-----------------------------------------------------------|---------|-----------------------|
| ▼ Aug 20, 2025 version files                              |         | 517.25 KB             |
| <a href="#">Charetteetal2024_CJFAS_Baselines_DOC.csv</a>  | Preview | 7.94 KB               |
| <a href="#">Charetteetal2024_CJFAS_Fish.csv</a>           | Preview | 491.01 KB             |
| <a href="#">Charetteetal2025.CJFAS.DOC.Fish.RScript.R</a> |         | 13.82 KB              |
| <a href="#">README.md</a>                                 |         | 4.48 KB               |
| Click names to download individual files                  |         | Download full dataset |

## Abstract

Freshwater ecosystems are affected by fluctuations in terrestrially derived dissolved organic carbon (DOC). Increased DOC runoff from watersheds can contribute to the "browning" of freshwater systems, altering water properties and disrupting food web dynamics, with potential

impacts on fish. This study examined how carbon sources and the trophic ecology of temperate freshwater fishes varied with DOC concentrations. We used stable isotope ratios of carbon ( $\delta^{13}\text{C}$ ) and nitrogen ( $\delta^{15}\text{N}$ ) to trace carbon sources in organisms at low trophic levels and to assess the trophic ecology of six fish species from 70 lakes in Sweden, Germany, Canada, and the United States. We found that both pelagic and benthic  $\delta^{13}\text{C}$  decreased as DOC increased, suggesting higher reliance on terrestrial carbon by lower trophic level organisms. Fish responses to elevated DOC were limited to certain species. Mid-trophic level European perch (*Perca fluviatilis*) increased pelagic diet proportions, while top predators like northern pike (*Esox Lucius*) and walleye (*Sander vitreus*) showed non-linear shifts. Except for walleye, no effect of DOC was detected on fish trophic position. Our findings suggest that DOC alters energy use and diet but not trophic position in some freshwater fish.

## README: Dissolved organic carbon enhances terrestrial carbon pathways with limited effects on freshwater fish trophic position



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### Description of the data and file structure

This supplementary dataset, essential to our research paper, provides data critical for interpreting our primary findings. As part of the Groupe de Recherche Interuniversitaire en Limnologie (GRIL) working group's investigation into the effects of browning on fish populations, this dataset integrates and expands upon findings from studies by Kelly et al. (2014), Tanentzap et al. (2014), Craig et al. (2015), Koizumi et al. (2018), Burke et al. (2020), Andersson (2021), Moslemi-Aqdam et al. (2021), and Johnston et al. (2024).

The dataset includes stable isotope ratios of carbon ( $\delta^{13}\text{C}$ ) and nitrogen ( $\delta^{15}\text{N}$ ) for pelagic and benthic baselines, along with fish samples across 70 lakes spanning Sweden (n=17), Germany (n=2), northern Ontario, Canada (n=35), Northwest Territories, Canada (n=10), Michigan, United States (n=1), and Alaska, United States (n=5). Target species include European perch (*Perca fluviatilis*), yellow perch (*Perca flavescens*), northern pike (*Esox lucius*), walleye (*Sander vitreus*), smallmouth bass (*Micropterus dolomieu*), largemouth bass (*Micropterus salmoides*), and ninespine stickleback (*Pungitius pungitius*). Additional parameters include data source, lake information, dissolved organic carbon concentrations (mg/L), date, the C/N ratios for fish samples, fish length (mm), pelagic diet proportions for individual fish—calculated using two-source mixing models from Post (2002)—and trophic positions, determined using both two-source and single-source mixing models from Post (2002).

## Files and variables

File: Charetteetal2024\_CJFAS\_Baselines\_DOC.csv

### Description:

#### *Variables*

- DataSource: Origin or author of the data
- Latitude: Latitude coordinate of the lake in decimal degrees
- Longitude: Longitude coordinate of the lake in decimal degrees
- Lake: Name of the lake studied
- Region: Geographic region where the lake is located
- Year: Year of data collection
- Month: Month of data collection
- Day: Day of data collection
- DOC\_mgL: Concentration of dissolved organic carbon in the lake (mg/L)
- Av\_pelagic\_d13C: Mean  $\delta^{13}\text{C}$  for the pelagic baseline in the lake
- Av\_pelagic\_d15N: Mean  $\delta^{15}\text{N}$  for the pelagic baseline in the lake
- Av\_benthic\_d13C: Mean  $\delta^{13}\text{C}$  for the benthic baseline in the lake
- Av\_benthic\_d15N: Mean  $\delta^{15}\text{N}$  for the benthic baseline in the lake
- d13C\_baselineDifference: Difference between pelagic and benthic  $\delta^{13}\text{C}$  baselines
- NA: Stands for Not Available. It is used in the dataset to indicate missing or unavailable information.

File: Charetteetal2024\_CJFAS\_Fish.csv

### Description:

#### *Variables*

- DataSource: Origin or author of the data
- Latitude: Latitude coordinate of the lake in decimal degrees
- Longitude: Longitude coordinate of the lake in decimal degrees
- Lake: Name of the lake studied
- Region: Geographic region where the lake is located

- Year: Year of data collection
- Month: Month of data collection
- Day: Day of data collection
- DOC\_mgL: Concentration of dissolved organic carbon in the lake (mg/L)
- Fish\_Species: Species of fish sampled
- SampleID: Unique identifier for each fish sample
- Length\_mm: Length of the fish sampled (in mm)
- d13C: Carbon isotope ratio ( $\delta^{13}\text{C}$ ) for fish
- d15N: Nitrogen isotope ratio ( $\delta^{15}\text{N}$ ) for fish
- C.N: Carbon-to-nitrogen ratio of fish
- Av\_pelagic\_d13C: Mean  $\delta^{13}\text{C}$  for the pelagic baseline in the lake
- Av\_pelagic\_d15N: Mean  $\delta^{15}\text{N}$  for the pelagic baseline in the lake
- Av\_benthic\_d13C: Mean  $\delta^{13}\text{C}$  for the benthic baseline in the lake
- Av\_benthic\_d15N: Mean  $\delta^{15}\text{N}$  for the benthic baseline in the lake
- d13Ccorr.lipids: Lipid-corrected  $\delta^{13}\text{C}$  value for fish using Kiljunen et al. (2006) equation.
- Pelagic diet proportion: Calculated pelagic diet proportion for each fish, based on two-source mixing models
- Trophic position: The Trophic position of each fish, calculated using two-source and one-source mixing models
- NA: Stands for Not Available. It is used in the dataset to indicate missing or unavailable information.

## Code/software

All analysis results presented in the paper can be exactly reproduced using the R Software. The R script is provided here: Charetteetal2025.CJFAS.DOC.Fish.RScript.R

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## Subject keywords

[Fish](#), [Freshwater ecosystems](#), [Stable isotopes](#), [Biological sciences](#), [browning](#), [Interspecific variation](#)

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